Kubernetes

Easy Understanding Concepts of Kubernetes:

Notes:

1. ***Kubernetes Basics:***
   - ***Pods****: Smallest deployable units in Kubernetes that can hold one or multiple containers.*
   - ***Nodes****: Worker machines in Kubernetes.*
   - ***Clusters****: A set of worker machines, called nodes, that run containerized applications.*
   - ***Kubectl****: Command-line tool for interacting with a Kubernetes cluster.*
   - ***Kubelet****: An agent running on each node in the cluster.*
   - ***API Server****: Front-end to the cluster's control plane.*
   - ***Etcd****: Consistent and highly-available key-value store used as Kubernetes' backing store for all cluster data.*
   - ***Control Plane****: The collection of processes that control Kubernetes nodes.*
   - ***Namespace****: Virtual clusters backed by the same physical cluster.*

2. ***Workloads & Controllers:***

   - ***Deployments****: Manages a replicated application.*
   - ***ReplicaSets****: Ensures that a specified number of replicas of a Pod are running at all times.*
   - ***StatefulSets****: Manages the deployment and scaling of a set of Pods, providing guarantees about ordering and uniqueness.*
   - ***DaemonSets****: Ensures that all (or some) nodes run a copy of a Pod.*
   - ***Jobs****: Creates one or more Pods and ensures that a specified number of them successfully terminate.*
   - ***CronJobs****: Manages time-based Jobs, such as running a Job at a specific time or periodic intervals.*
   - ***Horizontal Pod Autoscaler****: Automatically scales the number of Pods in a deployment, replica set, or replication controller based on observed CPU or memory usage.*

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3. ***Services & Networking:***

- ***Services***: *A way to expose an application running in Pods as a network service.*
- ***Ingress***: *Manages external access to services within a cluster.*
- ***Network Policies***: *Define how Pods communicate with each other.*
- ***Service Discovery***: *Mechanism to connect to services dynamically based on a logical name.*
- ***Load Balancer***: *A service that distributes network traffic across multiple Pods.*

4. ***Configuration & Secrets:***

- ***ConfigMaps***: *Manage configuration data separately from container images.*
- ***Secrets***: *Manages sensitive information, such as passwords, OAuth tokens, and ssh keys.*
- ***Environment Variables***: *Used within Kubernetes for service discovery.*
- ***Volumes***: *Persistent storage in Kubernetes.*
- ***Persistent Volume Claims (PVCs)***: *Request for storage by a user.*

5. ***Monitoring, Logging, and Debugging:***

- ***Kube-state-metrics***: *A simple service that listens to the Kubernetes API server and generates metrics about the state of the objects.*
- ***Prometheus***: *An open-source system monitoring and alerting toolkit.*
- ***Grafana***: *Open platform for beautiful analytics and monitoring (often used with Prometheus).*
- ***ELK Stack***: *Elasticsearch, Logstash, and Kibana used for logging in Kubernetes.*
- ***Kubernetes Dashboard***: *General-purpose web UI for Kubernetes clusters.*
- ***Kubectl Debug***: *Tool for debugging pods.*
- ***Heapster***: *Collects monitoring and performance metrics.*

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6. ***Security & Authorization:***

- ***Role-Based Access Control (RBAC)***: *Access control system.*
- ***Security Policies***: *Security constraints applied to a Pod.*
- ***Service Account***: *Identity attached to Pods to interact with the Kubernetes API Server.*
- ***Transport Layer Security (TLS)***: *Protocol for privacy and data integrity.*

7. ***Kubernetes Storage:***

- ***Persistent Volumes (PVs)***: *Offers storage to the cluster that is independent of Pod life cycles.*
- ***Storage Classes***: *Allow administrators to describe the "classes" of storage offered.*

8. ***Kubernetes Cluster Maintenance:***

- ***Node Maintenance***: *Taking a node down for service or replacing a failing node.*
- ***Cluster Upgrades***: *Upgrading the cluster to a newer version.*
- ***Backup and Disaster Recovery***: *Ensuring data continuity and integrity.*

9. ***Extending Kubernetes:***

- ***Custom Resources***: *Extension of the Kubernetes API.*
- ***API Server Extension***: *Custom endpoints in the Kubernetes API.*
- ***Custom Controllers***: *Automate handling of Custom Resources.*
- ***Operators***: *Application-specific controllers to create, configure, and manage instances of complex stateful applications.*

10. ***Advanced Features:***

- ***Service Mesh (e.g., Istio)***: *Manage microservices in a transparent way.*
- ***Pod Priority & Preemption***: *Specifies priorities for Pods and allows preemption of lower-priority Pods.*
- ***Taints and Tolerations***: *Allows a node to repel a set of Pods.*

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- ***Node Affinity***: *Controls where a Pod runs based on labels on nodes and conditions called node affinity rules.*
- ***Pod Presets***: *Injects information like secrets, volume mounts, and environment variables into pods at creation time.*

11. ***Kubernetes Cloud Integration:***

- ***Amazon EKS***: *Managed Kubernetes service on AWS.*
- ***Google Kubernetes Engine (GKE)***: *Managed Kubernetes service on Google Cloud.*
- ***Azure AKS***: *Managed Kubernetes service on Azure.*

12. ***Continuous Deployment/Integration in Kubernetes:***

- ***Jenkins***: *Popular open-source tool to perform continuous integration and build automation.*
- ***GitLab CI***: *Continuous integration service included with GitLab that builds and tests the software whenever a developer pushes code to the application.*

13. ***Helm: Kubernetes Package Manager:***

- ***Helm Charts***: *Packages of pre-configured Kubernetes resources.*
- ***Helm Repository***: *A collection of charts for Kubernetes packages.*

14. ***Kubernetes Custom Scheduling:***

- ***Custom Scheduler***: *You can implement a custom scheduler to have more control over the scheduling of your Pods.*

15. ***Kubernetes Federation:***

- ***Cluster Federation***: *Allows the synchronization of resources across multiple clusters.*

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16. ***Windows in Kubernetes:***

- ***Windows Nodes***: *Support for Windows worker nodes and Windows containers in Kubernetes.*

17. ***Kubernetes Testing:***

- ***Kubetest***: *Kubernetes end-to-end testing.*

18. ***Kubernetes Add-ons:***

- ***DNS***: *DNS server for service name resolution.*
- ***Web UI (Dashboard)***: *Web-based Kubernetes user interface.*
- ***Container Resource Monitoring***: *Time-series monitoring of resource utilization.*
- ***Cluster-level Logging***: *Save container logs to a logging backend.*

19. ***Other Essential Concepts:***

- ***Quotas***: *Constraints applied to resources like Pods, Persistent Volumes.*
- ***Annotations***: *Attach arbitrary non-identifying metadata to objects.*
- ***Labels and Selectors***: *Labels are key/value pairs attached to objects and selectors are used to select objects based on labels.*
- ***Liveness and Readiness Probes***: *Health checks for running applications.*

20. ***Kubernetes Failure Handling:***

- ***Pod Disruption Budgets (PDBs)***: *Provides constraints that limit voluntary disruptions for Pods.*

21. ***Kubernetes Architecture Concepts:***

- ***Cloud Controller Manager***: *Embeds cloud-specific control logic for the cloud provider.*
- ***Kube-Proxy***: *Maintains network rules and enables connection forwarding.*
- ***Container Runtime***: *The software used to run containers, e.g., Docker, containers.*

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22. ***Kubernetes Community and Development:***

- ***Special Interest Groups (SIGs)****: The Kubernetes project is divided into several Special Interest Groups or SIGs.*
- ***Contributing to Kubernetes****: Guide to becoming an active contributor to the Kubernetes project.*

23. ***Kubernetes Best Practices:***

- ***Logging Best Practices****: Guidelines for logging in Kubernetes.*
- ***Monitoring Best Practices****: Guidelines for monitoring in Kubernetes.*
- ***Security Best Practices****: Guidelines for securing your applications in Kubernetes.*
- ***Performance Best Practices****: Guidelines for ensuring optimal performance in Kubernetes.*

24. ***Kubernetes Automation and Autoscaling:***

- ***Cluster Autoscaler****: Automatically adjusts the size of the cluster, scaling it up or down as necessary.*
- ***Vertical Pod Autoscaler****: Automatically adjusts the amount of CPU and memory requested by containers in a Pod.*
- ***Horizontal Pod Autoscaler (HPA)****: Automatically scales the number of Pods in a deployment or replica set based on observed CPU or memory utilization.*

25. ***Kubernetes Development Tools:***

- ***Minikube****: Runs a single-node Kubernetes cluster inside a VM on your laptop for users looking to try out Kubernetes or develop with it day-to-day.*
- ***Skaffold****: Command line tool that facilitates continuous development for Kubernetes applications.*
- ***Kompose****: Conversion tool for all Docker Compose users to help them move to Kubernetes.*
- ***Kubeadm****: Tool for bootstrapping a best-practice Kubernetes cluster.*

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26. *Advanced Kubernetes Networking:*

- ***Network Plugins**: Extend Kubernetes networking.*
- ***CNI (Container Network Interface)**: Standard for writing plugins to configure network interfaces in Linux containers.*
- ***Flannel**: Overlay network provider.*
- ***Calico**: Provides secure network connectivity.*

27. *Kubernetes Performance Tuning:*

- ***Kubelet Garbage Collection**: Cleanup of unused images or containers.*
- ***Kernel Tuning**: Adjusting Linux kernel parameters to optimize performance.*

28. *Kubernetes Service Mesh:*

- ***Istio**: Open platform to connect, manage, and secure microservices.*
- ***Linkerd**: Another popular service mesh for Kubernetes.*

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Kubernetes Commands & Usages:

1- *Basic Cluster Information:*

- *kubectl cluster-info: Display information about the cluster.*
- *kubectl version: Display version info.*

2- *Working with Nodes and Cluster:*

- *kubectl get nodes: List nodes in a cluster.*
- *kubectl describe node <node-name>: Show details of a specific node.*

3- *Working with Pods:*

- *kubectl get pods: List all pods in all namespaces.*
- *kubectl run <name> --image=<image>: Deploy a new Pod with a given image.*
- *kubectl describe pod <pod-name>: Describe a specific pod.*
- *kubectl logs <pod-name>: Fetch the logs from a pod.*
- *kubectl delete pod <pod-name>: Delete a specific pod.*

4- *Working with Deployments:*

- *kubectl create deployment <name> --image=<image>: Create a new deployment.*
- *kubectl get deployments: List all deployments.*
- *kubectl describe deployment <deployment-name>: Describe a specific deployment.*
- *kubectl scale deployment <deployment-name> --replicas=<num>: Scale up/down a deployment.*

5- *Working with Services:*

- *kubectl expose deployment <name> --type=LoadBalancer --port=8080: Expose a deployment as a service.*
- *kubectl get services: List all services.*
- *kubectl describe service <service-name>: Describe a specific service.*

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6- *Config and Storage:*

- *kubectl get configmaps: List all config maps.*
- *kubectl create configmap <name> --from-file=<path>: Create a config map from a file.*
- *kubectl get secrets: List all secrets.*
- *kubectl create secret: Create a secret.*
- *kubectl get pv: List all persistent volumes.*
- *kubectl get pvc: List all persistent volume claims.*

7- *Namespaces and Context:*

- *kubectl get namespaces: List all namespaces.*
- *kubectl config get-contexts: Show all contexts.*
- *kubectl config use-context <context-name>: Switch to a different context.*

8- *Others:*

- *kubectl apply -f <filename>: Apply a configuration from a file.*
- *kubectl delete -f <filename>: Delete resources defined in a file.*
- *kubectl exec -it <pod-name> -- /bin/sh: Execute a command inside a running pod.*
- *kubectl port-forward <pod-name> <local-port>:<pod-port>: Forward a port from a running pod to a local port.*

9- *Advanced:*

- *kubectl get all: List all resources.*
- *kubectl rollout status deployment/<deployment-name>: View the rollout status of a deployment.*
- *kubectl rollout history deployment/<deployment-name>: View the history of a deployment.*
- *kubectl rollout undo deployment/<deployment-name>: Rollback to a previous version of a deployment.*

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10- *Monitoring & Logging:*

- *kubectl top nodes: Display resource (CPU/Memory/Storage) usage of nodes.*
- *kubectl top pods: Display resource (CPU/Memory/Storage) usage of pods.*

11- *Autoscaling:*

- *kubectl autoscale deployment <deployment-name> --min=<min-pods> --max=<max-pods> --cpu-percent=<cpu-util-percentage>: Auto scale a deployment based on CPU utilization.*

12- *Working with Helm:*

- *helm list: List releases.*
- *helm install <chart>: Install a helm chart.*
- *helm uninstall <release-name>: Uninstall a helm release.*

13- *Using Network Policies:*

- *kubectl get networkpolicies: List all network policies.*
- *kubectl describe networkpolicy <policy-name>: Describe a specific network policy.*

14- *Using CronJobs and Jobs:*

- *kubectl get cronjobs: List all cronjobs.*
- *kubectl get jobs: List all jobs.*
- *kubectl logs job/<job-name>: Fetch logs from a job.*

15- *Working with RBAC:*

- *kubectl get roles: List all roles in the current namespace.*
- *kubectl get clusterroles: List all cluster roles.*

16- *Using kubectl Plugins:*

- *kubectl krew search: Search plugins available for kubectl.*
- *kubectl krew install <plugin-name>: Install a kubectl plugin.*

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